Cod: fewest-tap routes from an image to a floating PiP window — research (Oct 15, 2026, ~6 PM PDT)

Scope: read-only research for josh. Nothing posted, bought or changed. Code facts come from the read-only copy of github.com/Triwix/Cod at files/apple-forum/cod-src (commit 1764945). Every claim has a URL; anything I could not verify is marked **unverified**. "Taps" count what the user does after the image is on screen (share-sheet scrolling not counted; a swipe counts as a gesture, noted separately).

TL;DR ranking (fewest taps × most likely to pass review)

#	Route	Taps after the image is on screen	Supported by Apple?	Auto-PiP once Cod is up?	Verdict
1	**Share → Cod share extension → responder-chain `open(_:options:completionHandler:)` → Cod foregrounds → PiP** (today's Fast build)	2 (Share, Cod)	**No.** Apple engineer: "no supported way… except Today and Widgets" (thread 773342). Works on iOS 26/27 per thread 848791 and SO; one SO commenter says it failed for them on iOS 26 (keyboard ext.)	Yes (Cod already does it)	Fewest ta unsuppor documen rejection trick four no Apple approval either
2	**Control Center / Lock Screen / Action button control → `OpenIntent` → Cod foregrounds → "Float latest screenshot" via PhotoKit → PiP**	1 tap + 1 swipe (Control Center), 1 tap (Lock Screen control), 1 press (Action button)	**Yes** — documented API (WidgetKit controls, iOS 18+)	Yes: app launches active, Cod's `FloatLatestScreenshotIntent` path already exists	**Best supporte route fo screensh case.** No Cod yet (`Control` in the rep
3	**Back Tap (double/triple-tap the phone's back) → shortcut → Cod App Intent (`.foreground(.immediate)`) → PiP**	0 taps (a gesture)	Yes (Accessibility feature + App Intents)	Yes	Great for screensh needs on

#	Route	Taps after the image is on screen	Supported by Apple?	Auto-PiP once Cod is up?	Verdict
					setup in Settings
4	**Share → "Float with Cod" shortcut (Show in Share Sheet) → runs Cod's App Intent with the image (`IntentFile`) → Cod foregrounds → PiP**	2 (Share, the shortcut) — same count as route 1	Yes (Shortcuts' documented share-sheet mechanism)	Yes	**Support twin of r 1**; cost is time shor install (yo ship it as iCloud sh link and p once)
5	Share → Cod share extension saves to App Group → local notification "Tap to float" → Cod → PiP	3 (Share, Cod, notification)	Yes — the pattern Apple-compliant devs fall back to (SO answer; forum 719905)	Yes	Fallback w isn't acce +1 tap ar
6	Share → Cod share extension starts PiP **inside the extension** (no app)	2	**Unverified / almost certainly not workable**: no doc or shipping app found that does it; PiP needs the app's audio background mode, which an extension cannot declare	n/a	Don't bui
7	Share → Cod extension completes; user opens Cod manually (today's default build)	3+ (Share, Cod, Home/ Cod icon)	Yes	—	Baseline

Live Activities and widgets are 1-tap launchers too (see §4) but add nothing over 2 for this use.

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1. Can a Share or Action extension start PiP itself, without the app?

Short answer: nothing documented says yes; everything around it says no. Treat as not possible.

- `AVPictureInPictureController` carries **no `NS_EXTENSION_UNAVAILABLE` annotation** in the iOS 18.0 SDK header (`API_AVAILABLE(ios(9.0)…)` only), so the class compiles under `APPLICATION_EXTENSION_API_ONLY`. Header: https://raw.githubusercontent.com/xybp888/iOS-SDKs/master/iPhoneOS18.0.sdk/System/Library/Frameworks/AVKit.framework/Headers/AVPictureInPictureController.h — compiling is not the same as working (see next bullets). I did not check the iOS 26/27 SDK headers (not mirrored where I could reach them) — **unverified** for those SDKs.
- Apple's AVPictureInPictureController docs: "To use Picture in Picture, you need to configure your app to support background audio playback" and "check the isPictureInPicturePossible property value to determine whether PiP is possible in the current context." https://developer.apple.com/documentation/avkit/avpictureinpicturecontroller — the audio background mode is an app Info.plist capability; share/action extensions have no `UIBackgroundModes`, and Apple's extension guide says an extension "typically terminates soon after it completes the request." https://developer.apple.com/library/archive/documentation/General/Conceptual/ExtensibilityPG/ExtensionOverview.html
- The extension's view is a remote view hosted by Photos ("The extension displays its view within the context of the host app", same guide). PiP is a system window owned by the process that plays; an extension that is torn down when it calls `completeRequest` cannot keep a player alive. **Unverified by a direct Apple statement** — this is inference from the two docs above plus the absence of any shipping example.
- Shipping-app evidence: every App Store app that floats share-sheet content through PiP that I found opens its own app first ("Picture in Picture with PiP-it! … operates as a separate app and doesn't integrate directly to the YouTube app", App Store review, 2021: https://apps.apple.com/us/app/pip-it-picture-in-picture/id1513646283). PiPifier/CornerTube-style Safari action extensions start PiP by running JavaScript that triggers **Safari's own** video PiP, which does not apply to Photos or to still images. https://apps.apple.com/us/app/pipifier/id1234771095 , https://github.com/arnoappenzeller/PiPifier
- AVKit/AVPlayerViewController in extensions: no "unavailable in extensions" marking in docs (https://developer.apple.com/documentation/avkit/avpictureinpicturecontroller), but same runtime problem. No iOS 26 or iOS 27 change to any of this appears in WWDC25/26 AVKit or App Intents material I read (WWDC26 session 345 lists ValueRepresentation, RelevantEntities, EntityCollection, SyncableEntity, union values, LongRunningIntent, ExecutionTargets — nothing about PiP or extensions launching apps: https://developer.apple.com/videos/play/wwdc2026/345/).

2. Supported ways to get Share sheet → Cod in the foreground automatically

2a. What Apple says about extensions opening their app

- `NSExtensionContext.open(_:completionHandler:)`: "Each extension point determines whether to support this method… In iOS, the Today and iMessage app extension points support this method." https://developer.apple.com/documentation/foundation/nsextensioncontext/open(_:completionhandler:) — i.e. **not Share, not Action** (same for both extension types). The App Extension Programming Guide: "A Today widget (and no other app extension type) can ask the system to open its containing app." https://developer.apple.com/library/archive/documentation/General/Conceptual/ExtensibilityPG/ExtensionOverview.html
- Apple Frameworks Engineer, Jan 2025 (thread 773342, marked the accepted answer): "There's no supported way for you to launch your app directly from App Extensions, except Today and Widgets (which requires OpenURLIntent and is available to processes that can use App Intents). with the APIs currently available. If you'd like us to consider adding the necessary functionality, please file an enhancement request using Feedback Assistant." https://developer.apple.com/forums/thread/773342 (read via the browser Oct 15, 2026). Same engineer gave the identical answer for ShieldAction extensions, Nov 2022 and Jan 2025: https://developer.apple.com/forums/thread/719905
- Thread 848791 (the near-identical responder-chain question): as of Oct 15, 2026, **0 replies, 133 views, 1 participant**, created ~2 weeks ago; the poster reports the technique "launches the containing app and the completion handler reports success == true, on both iOS 26 and iOS 27" and that `NSExtensionContext.open` reports `success == false` from a share extension. https://developer.apple.com/forums/thread/848791
- **Action extension vs Share extension**: identical rules. Action extensions (`com.apple.ui-services` / `com.apple.services`) are hosted the same way and get no `open` support either (doc above). A May 2025 reply in 773342 asks exactly this about competitors' action extensions that "immediately cold start their app" — unanswered by Apple. https://developer.apple.com/forums/thread/773342

2b. App Intents: foreground modes (the supported "open my app" mechanism)

- `openAppWhenRun` is **deprecated in iOS 26** ("Use supportedModes instead") and "Setting this property to true generates an error if the app intent runs in an app extension." https://developer.apple.com/documentation/appintents/appintent/openappwhenrun
- `supportedModes` (iOS 26+): `.foreground(.immediate)` "bring the app to the foreground before the action runs"; `.foreground(.dynamic)` run in background and optionally transition; `.foreground(.deferred)` transition before completion. https://developer.apple.com/documentation/appintents/appintent/supportedmodes , https://developer.apple.com/documentation/appintents/intentmodes

- `ForegroundContinuableIntent` (iOS 16.4–25) is deprecated in iOS 26 in favour of `.foreground(.dynamic)`. https://developer.apple.com/documentation/appintents/foregroundcontinuableintent
- `OpenIntent`: "The system automatically brings your app to the foreground to run this app intent." https://developer.apple.com/documentation/appintents/openintent
- **Cod already has these**: `FloatWithCodIntent` (`@Parameter var image: IntentFile`, `openAppWhenRun = true`, `supportedModes { .foreground(.immediate) }`) and `FloatLatestScreenshotIntent`, plus an `AppShortcutsProvider` — cod-src/Cod/Intents/CodIntents.swift lines 98–150, 171+. `PhotoLibraryService.latestScreenshot()` fetches `PHAssetMediaSubtype.photoScreenshot` (cod-src/Cod/Capture/PhotoLibraryService.swift: 127–151).

2c. Can an App Intent / App Shortcut appear directly in the share sheet?

- **No API found.** The share sheet's third-party rows are share extensions, action extensions, and user shortcuts with "Show in Share Sheet". App Intents surface in Siri, Spotlight, Shortcuts, widgets, Controls, the Action button and Apple Pencil — Apple's own list in WWDC25 "Get to know App Intents" (https://developer.apple.com/videos/play/wwdc2025/244/) and the App Intents overview (https://developer.apple.com/documentation/appintents/getting-started-with-the-app-intents-framework) — and none of the WWDC24/25/26 App Intents sessions mentions the share sheet. An `IntentFile` parameter lets a shortcut hand share-sheet input to your intent; it does not list the intent in the sheet by itself. **Negative claim — unverified beyond "not found in Apple's docs/sessions".**
- A June 2026 forum thread (iOS 27) asks for a supported way for a third-party app to receive another app's on-screen image via Siri/App Intents; the poster tried `IntentFile`, `.photos.setFilter` schema and `OpenIntent`, none route from Photos; 4 community replies, **no Apple reply** as of Oct 15, 2026. https://developer.apple.com/forums/thread/836167

2d. Shortcuts "Show in Share Sheet" (supported, one-time setup)

- Apple: "To allow a shortcut to run from within other apps, you must enable it to appear in the share sheet… turn on Show in Share Sheet… A new action appears at the beginning of your workflow defining inputs the shortcut will accept." https://support.apple.com/guide/shortcuts/run-a-shortcut-from-another-app-apd163eb9f95/ios
- Shortcuts' own share actions include "Open App" and "Open URLs", so a share-sheet shortcut is allowed to bring an app forward. https://support.apple.com/guide/shortcuts/share-actions-apdaf74d75a5/ios
- Flow: Photos → Share → "Float with Cod" (shortcut row) → the shortcut runs `FloatWithCodIntent` with the image as `IntentFile` → `.foreground(.immediate)` brings Cod up → PiP. Taps: 2 (same as the extension). Whether iOS shows a one-time "Allow 'Float with Cod' to open Cod?" confirmation — **unverified on device**.
- Install friction: a shortcut can be published as an iCloud link and offered from inside Cod (one tap to add). Cod's App Shortcuts (CodShortcuts) already exist; App Shortcuts

themselves have no "Show in Share Sheet" switch (that lives on user-created shortcuts) — **unverified**: I did not find Apple text either way for App Shortcuts in the sheet.

2e. Does `NSExtensionContext.open` work from Share/Action extensions on current iOS?

- No. Documented for Today and iMessage only (link above); the 848791 poster observes `success == false` on iOS 26/27; the 773342 poster was told it "may require additional review"; 719905's May 2026 poster: "extensionContext.open(_:) fails silently in this extension context." https://developer.apple.com/forums/thread/719905

3. Shipping apps that do Share → app foreground, and how

- **Many do**, by the responder-chain trick. The highest-voted current Stack Overflow answer (Sep 2024, iOS 18 SDK): the old `perform(#selector(openURL:))` form now logs "BUG IN CLIENT OF UIKIT: The caller of UIApplication.openURL(:) needs to migrate to… open(:options:completionHandler:). Force returning false", and the fix is the `as? UIApplication` + `open(_:options:completionHandler:)` form Cod uses. https://stackoverflow.com/a/78975759 Comments under it: "Tested and confirmed with a share extension on iOS 18" (Sep 23, 2024); "This works for iOS18 and below but it failed on iOS26" (Sep 17, 2025) and "specially when keyboard does not have full access this does not work in ios26" (Oct 6, 2025, keyboard extension); "works on iOS26.4" (May 27, 2026). So: **works for most on iOS 26.x; at least one report of failure on early iOS 26; keyboard extensions differ.**
- A Jan 2025 answer on the same question (matt neuburg, score 6) describes the full recipe (invisible extension + `NSExtensionActionWantsFullScreenPresentation` + responder chain) and says: "many apps do in fact have share extensions that do this… do not actually do what I'm about to describe! It's illegal and Apple doesn't want you to do it." https://stackoverflow.com/a/79369242
- Forum members in 773342: "There are apps on the store that do this… This could result in app review issues - if they discovered it." Two Screen Time apps (App Lock id6448239603, AppLocker id1132845904) and one-sec are named in 719905 as opening the parent app from a shield extension. https://developer.apple.com/forums/thread/773342 , https://developer.apple.com/forums/thread/719905
- Reddit r/iOSProgramming (May 2023), from the search snippet (Reddit blocks this box): "I know the workarounds on how to launch the app from share extension. But how is apple allowing them to do it without rejection?" and a reply: "now I am getting errors that openURL is no longer supported from ShareExt." https://www.reddit.com/r/iOSProgramming/comments/137ratu/ — **snippet only, thread body unverified.**
- Named consumer apps whose share extension opens the host app (e.g. Instagram/WhatsApp): widely believed but **I could not verify their mechanism from any source — unverified; left out of the recommendation.**
- **Rejections found**: one documented, May 2026, under 2.5.1 — but for the private `LSApplicationWorkspace`, detected by static analysis, not for the responder chain: https://developer.apple.com/forums/thread/719905 (post "May 2026 rejection for

LSApplicationWorkspace"). The same poster calls the responder-chain selector trick "documented broken on iOS 18+" (true for the `openURL:` selector, superseded by the `open(_:options:)` form) and names the compliant fallback: "local notification with a deep link — works, but adds ~1 second and an extra tap." **No report of a rejection for the `as? UIApplication` + `open(_:options:completionHandler:)` form was found** on the forums, Stack Overflow or search; equally, **no Apple statement approving it** exists, and thread 848791 asking exactly that is unanswered. Static analysis cannot flag it (public API, public symbols), so the exposure is a reviewer noticing the behaviour, or a future iOS build closing it (the Sep 2025 iOS 26 report shows that can happen without notice).

- App Review Guidelines 2.5.1 ("Apps may only use public APIs and must run on the currently shipping OS… use APIs and frameworks for their intended purposes") and 2.5.4 ("Multitasking apps may only use background services for their intended purposes: VoIP, audio playback, location, task completion, local notifications, etc.") https://developer.apple.com/app-store/review/guidelines/

4. Other low-tap triggers for "float the latest screenshot / photo"

All of these are supported APIs; all end with Cod active in the foreground, where `startPictureInPicture()` can run immediately (Cod already does this in `PiPPinController`, with `canStartPictureInPictureAutomaticallyFromInline = false`, cod-src/Cod/Pins/PiPPinController.swift:331,559). PiP needs the app's player layer on screen in an active scene — Apple: check `isPictureInPicturePossible` "in the current context" (https://developer.apple.com/documentation/avkit/avpictureinpicturecontroller). `canStartPictureInPictureAutomaticallyFromInline = true` is the other way in: PiP "starts automatically when the controller embeds its content inline and the app transitions to the background" (https://developer.apple.com/documentation/avkit/avpictureinpicturecontroller/canstartpictureinpictureautomaticallyfrominline) — i.e. the user swipes Home and the picture floats, no tap; Apple says only set it for "content that you intend to be the user's primary focus", which a float-this-image app can argue.

Trigger	Taps	Mechanism	Source
Control Center control (iOS 18+)	swipe down + 1 tap	`ControlWidgetButton` whose action is an `OpenIntent` ("launch your app to a specific view… from Control Center, the Lock Screen, or by using the Action button"); run `FloatLatestScreenshotIntent` on launch	https://developer.apple.com/documentation/widgetkit/creating-controls-to-perform-actions-across-the-system
Lock Screen control	1 tap (+ Face ID)	same control, placed in a Lock Screen slot	same
Action button	1 press	assign the control or the App Shortcut ("App Shortcuts… can be configured	https://developer.apple.com/videos/play/wwdc2025/244/ , controls doc above

Trigger	Taps	Mechanism	Source
(iPhone 15 Pro and later)		to run from the Action Button or Apple Pencil", WWDC25)	
Back Tap	0 taps (double/triple tap the back)	Settings > Accessibility > Touch > Back Tap → a shortcut that runs Cod's intent	https://support.apple.com/guide/shortcuts/run-shortcuts-tapping-iphone-apd897693606/ios
Home Screen widget / App Shortcut tile	1 tap (from Home)	widget `Link` / `widgetURL` or an interactive button with an `OpenIntent`; buttons that don't open the app run in the background (not useful for PiP)	https://developer.apple.com/documentation/widgetkit/adding-interactivity-to-widgets-and-live-activities
Live Activity (Dynamic Island / Lock Screen)	1 tap	tapping the activity deep-links into the app; buttons inside run intents without launching the app, so a tap on the activity body is the one that works; activities expire (community: ~8 h Dynamic Island / 12 h Lock Screen — **unverified against Apple's page**, which did not render for me)	https://developer.apple.com/documentation/activitykit/displaying-live-data-with-live-activities
Spotlight / Siri App Shortcut ("Float my latest screenshot in Cod")	1 tap or voice	existing CodShortcuts	https://developer.apple.com/documentation/appintents/acceleratingappinteractionswithappintents

Note on the screenshot case specifically: Cod's `RETURN_RESEARCH.md` (Oct 6, 2026 tests on iOS 26.7.1) already found no way to return beneath the screenshot preview; a Control/Action-button/Back-Tap launch after the screenshot thumbnail disappears sidesteps that entirely, and `latestScreenshot()` picks the right asset.

5. Competitors: still-image PiP apps that are live on the App Store today

PiP-for-still-images is clearly being approved and kept live (all verified via the App Store lookup/search API on Oct 15, 2026):

App	Still images via PiP?	Last update	Min iOS	Link
	Yes: "Picture-in-Picture for images and videos from your Photos, Camera, and		17.0	

App	Still images via PiP?	Last update	Min iOS	Link
PiP – Picture in Picture (Supagarn Pattananuchart)	Files app"; a review (Dec 2023) praises "opening image directly from photos app"; another: "Works with pics (no conversion to video)"	Sep 20, 2026 (v3.0)		https://apps.apple.com/us/app/pip-picture-in-picture/id1635796246
Magic PiP (Jinyu Meng)	Yes: "Multiple Images… Allow open all the stuff from the share sheet"	Aug 26, 2025 (v3.8.0)	15.0	https://apps.apple.com/us/app/magic-pip/id1613046504
Pip Up: Persist/Float Post-It (Kunliang Wu)	Yes: "Select any local photo/image and paste it… always stays on top"	Sep 15, 2025	17.0	https://apps.apple.com/us/app/pip-up-persist-float-post-it/id6483210322
Always On Top – PDF/Web/Images (same dev)	Yes	Sep 15, 2025	18.0	https://apps.apple.com/us/app/always-on-top-pdf-web-images/id6747107679
Sightline: Refs PiP & Trace	Yes: "Float images and video references above your art app"	Aug 30, 2026 (v1.06)	**26.0**	https://apps.apple.com/us/app/sightline-refs-pip-trace/id6758032594
Split Screen – Floating Apps (1,092 ratings) / PiP Splitware	Yes: "Display Photos, Videos on Media Mode"	Sep 20, 2026	15.0	https://apps.apple.com/us/app/split-screen-floating-apps/id6499024218
SuperX – PIP Extension	Yes: "Support Local File: Video, Image, PDF"; release notes "Adapted for iOS 27"	Oct 14, 2026 (v3.1)	17.0	https://apps.apple.com/us/app/superx-pip-extension/id6450839791
PiP-it! (Artamata)	Video only, but its share-sheet → PiP flow is the closest UX analogue ("Share > More > Picture in Picture with PiP-it! and boom"); release notes "Updates for iOS 26"	Feb 2, 2026 (v2.7)	18.0	https://apps.apple.com/us/app/pip-it-picture-in-picture/id1513646283

So the 2.5.4 / "PiP for a still image" worry is small: at least seven live apps do it, several updated in the last month and one built for iOS 26. How each gets from the share sheet into its app (responder chain vs. something else) is **not verifiable from the outside — unverified**; none documents it, and every one I checked has the user land in the app before the window floats.

Snapper (the target): Snapper 3 on Havoc — "Pin cropped screenshots to display information across applications" — https://havoc.app/package/snapper3 , https://www.idownloadblog.com/2023/06/27/snapper-3/

Recommendation

Keep the share extension (it is the discoverable entry point users expect) but make its launch conformant and add one supported 1-tap path: ship a **Control Center / Lock Screen / Action-button control** backed by an `OpenIntent` that runs `FloatLatestScreenshotIntent` (one tap + a swipe, or one press, fully documented API, and it dodges the screenshot-preview return problem Cod already hit), and offer a **"Float with Cod" shortcut** (Show in Share Sheet → `FloatWithCodIntent` with the image as `IntentFile`) via an iCloud link so Photos → Share → Float with Cod → PiP is two taps through Apple's own mechanism. For the share extension itself, the compliant behaviour is save-to-App-Group + a local notification with a deep link (three taps, Apple-blessed); the responder-chain `open(_:options:completionHandler:)` launch works on iOS 26/27 and no rejection of that exact form is on record, but Apple has said twice (Jan 2025, thread 773342; ShieldAction thread 719905) there is "no supported way" to launch an app from an extension, thread 848791 asking precisely your question is unanswered after two weeks, and one SO report shows it breaking on an iOS 26 build — so treat it as a bonus you can switch off with a flag, not the product's spine, and file the enhancement request Apple keeps asking for (then quote the FB number in your forum post). PiP for a still image is not the risk: seven live App Store apps do exactly that, one of them iOS 26-only.